CPE 231 – Algorithms (1/69)

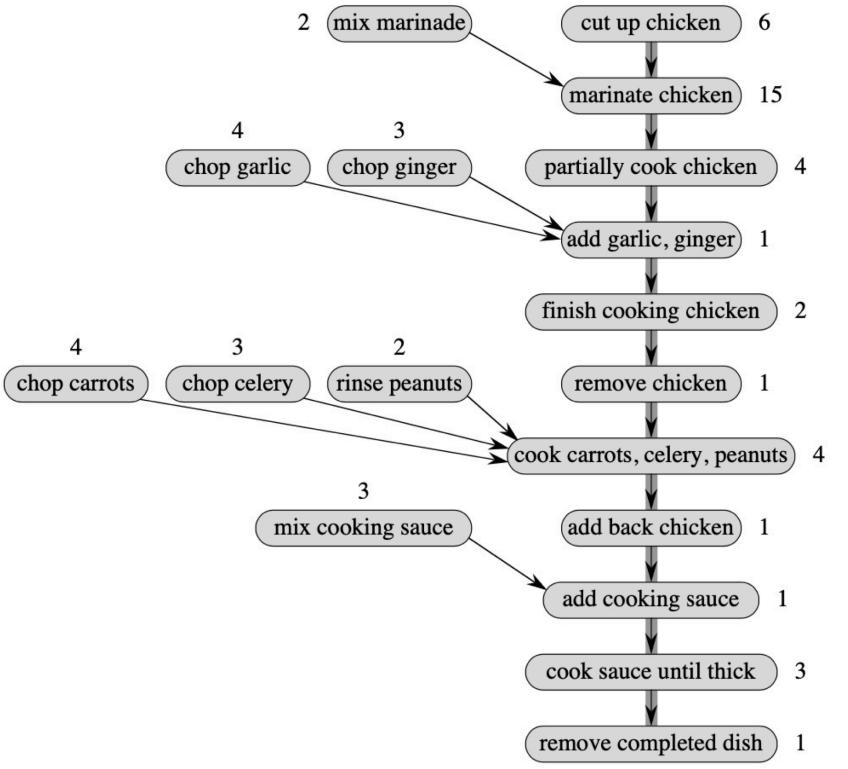
Homework II (Due by Sept. 7, 2026)

1. (10 points) Quickselect

   (a) (5 points) Given an input array $A[0 \ldots N-1]$, write pseudocode for a non-recursive version of Quickselect based on Lomuto partitioning.

   (b) (5 points) Apply the algorithm in (a) to find the median of the list of numbers `9, 12, 5, 17, 20, 30, 8`. Trace your algorithm by showing how the search segment has changed in each iteration to verify its correctness.

2. (10 points) From the steps for making Kung Pao chicken below, determine the quickest possible time we can finish it by using the longest path DAG algorithm. Draw the input graph G for the algorithm and show how the distance labels d_v are updated in each iteration.



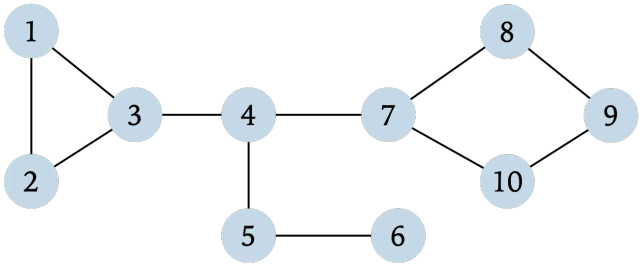
3. (20 points) Write a C program for DFS traversal on a unigraph.

   ▷ The program accepts an input file containing the number of nodes N in the first line and a list of node pair (i, j) as edges in the following lines.

   ▷ The node labels must be between 1 to N.

   ▷ Use the input to construct an adjacency list for the traversal. Then, print out the visit order and the finish order from the traversal.

   Submit the source code and the output of your program for the unigraph below:



4. (20 points) Modify your program in the previous question to implement Tarjan's algorithm to find articulation points in a graph. Submit the source code and the output of your program for the unigraph above.